

Question:

Thank you for this earlier analysis. Could you please estimate the average power of the photon flux that you estimated earlier? (The range, of course).

Response:

Using $E_\gamma \approx 386$ eV per photon:

Scenario	Flux	Average power
Pessimistic	1×10^{13} ph/s	~0.6 mW
Central estimate	5×10^{14} ph/s	~31 mW
Optimistic	1×10^{17} ph/s	~6.2 W

So the average power spans roughly **0.6 mW to ~6 W**, with ~30 mW as the central guess — again driven mainly by the same uncertainties (ring current, laser cavity finesse, IP overlap). For context, 30 mW of 386 eV X-rays would be a respectable lab-scale soft X-ray source, while 6 W would be exceptional and approach the output of much larger synchrotron beamlines in this energy range.

Question:

Could you confirm my guess that this design approach would not be suitable for creating a EUV source of 13.5 nm, since such sources require hundreds of Watts average power?

Response:

Confirmed, broadly — and the gap is even larger than the power numbers alone suggest.

Power mismatch: At 13.5 nm, photon energy is ~92 eV (vs 386 eV at 3 nm). For the same photon flux, that's already ~4× less power. Even at the optimistic end (~6 W at 3 nm), you'd be looking at ~1.4 W at 13.5 nm — still 2 orders of magnitude short of the "hundreds of watts" benchmark that EUV lithography light sources target.

But the deeper problem is kinematic, not just a power scaling issue. To shift the Compton output from 3 nm to 13.5 nm with a 600 nm drive laser, you'd need to roughly halve γ — i.e., drop the electron energy to around 1.5 MeV. At lower γ :

- The Compton cross-section scaling ($4\gamma^2$) means each photon-electron interaction produces ~4× fewer high-energy photons for the same N_e , N_{ph} , geometry
- Lower-energy electron rings are even harder to push to high current (more space-charge, more IBS, shorter lifetime)

- The "water window" target (2.3–4.4 nm) is precisely what motivated the 3 MeV design point — it's not an arbitrary choice, it's matched to this electron energy and 600 nm laser

So this concept is purpose-built around the water-window regime. Reaching 13.5 nm would require either a different electron energy regime entirely, a different drive laser wavelength, or fundamentally different photon multiplication (e.g., higher harmonics, different scattering geometry) — at which point you're really designing a different source, not retuning this one.

For EUV lithography power levels, the dominant technologies remain laser-produced plasma (LPP, tin droplets) and free-electron lasers, both of which solve the photon-flux problem through entirely different mechanisms (broadband plasma emission or coherent FEL gain) rather than incoherent Compton scattering from a stored beam.

Bottom line: your intuition is right — this is a water-window source concept, and EUV-at-power is a different design problem.

Question:

Agree! Could we alternatively consider pushing this design from water window into more harder X rays? Perhaps 100 keV photon energy? We can consider increasing the energy of the SRF section for that. Once you would estimate parameters for this design, could you again consider EUV approach, but when 13.5 nm photons are generated not via Compton, but with wiggler radiation (incoherent, not FEL mechanism). What energy of electrons is realistically needed to achieve such 13.5 nm with practically possible wigglers, and what average power could be possible?

Response:

Both extensions can be evaluated. Here's the summary:

1. Pushing to ~100 keV hard X-rays (increase SRF energy)

To shift the Compton output from 3 nm (386 eV) to **100 keV**, you need $\gamma \approx 110$, i.e. electron kinetic energy $\approx$ **56 MeV** (vs 3 MeV). Photon energy scales roughly as γ^2 , so a $\sim 16\times$ increase in γ gives $\sim 256\times$ higher photon energy — consistent with 386 eV $\rightarrow$ ~ 99 keV.

Implications:

- If the same SRF cell gradient applies (~ 1 MeV/cell as in the original 3-cell, 3 MeV design), you'd need **~ 56 cells** instead of 3 — a major cryomodule scale-up, not a small tweak.
- Magnetic rigidity grows to $B\rho \approx 0.19$ T·m, still compact (bend radius ~ 12 cm at 1.5 T). Synchrotron radiation loss per turn is negligible (~ 7 eV/turn) — not a constraint at this energy.

- If the same flux ($\sim 5 \times 10^{14}$ ph/s) somehow carried over, power at 100 keV would be **~ 8 W** (central), spanning **~ 0.16 mW – 1.6 kW** across the same flux uncertainty range.
- But this "same flux" assumption is optimistic: a 56 MeV ring is much harder to fill to Ampere-scale via LWFA top-up. LWFA naturally delivers MeV-scale electrons; reaching 56 MeV means either a much more powerful LWFA stage or substantial post-acceleration. This is really a different, larger machine — not a simple retune.

2. 13.5 nm EUV via incoherent wiggler radiation

Using the standard critical-wavelength relation ($\lambda_c \approx 1.864/(B \cdot E^2[\text{GeV}])$):

B field Required electron energy

0.5 T ~ 525 MeV

1.0 T ~ 372 MeV

1.5 T ~ 303 MeV

2.0 T ~ 263 MeV

So **~ 250 – 550 MeV**, roughly two orders of magnitude above the original 3 MeV concept — wigglers don't help reach 13.5 nm cheaply in energy terms; you genuinely need a few-hundred-MeV ring.

Power: with a modest 50-period, 2.5 m wiggler at $B \approx 1$ – 1.5 T:

- At 100 mA: total SR power ~ 25 W, with $\sim 2\%$ in a useful EUV bandwidth $\rightarrow$ **~ 0.5 W**
- At 0.5–1 A: total power ~ 125 – 250 W, in-band $\rightarrow$ **~ 2.5 – 5 W**

So even at very optimistic (Ampere-scale) currents, incoherent wiggler radiation gives **single-digit watts** in a useful EUV band — not the "hundreds of watts" that EUV lithography sources require. That power level is essentially only reachable via LPP or FEL mechanisms, confirming your original instinct held even after switching from Compton to wiggler radiation.

Bottom line: the hard X-ray (100 keV) direction is at least dimensionally plausible as an extension of this concept (with major SRF scale-up), while the wiggler-EUV route, though it points to a totally different (much higher-energy) machine, still falls short of lithography-class power by roughly 2 orders of magnitude.

Question:

Thank you. Coming back to initial design optimized for 13.5nm EUV, let's discuss LWFA injection part. The most recent record energy coming from LWFA are around 10 GeV. But in our case, we need to optimize LWFA to a very low energy of a fraction of MeV. Is this feasible? Could you please look at very first results from LWFA (which were at much lower energy) and suggest plausible parameters of plasma and laser, to create the LWFA injector for our source?

Response:

Yes — feasible, and arguably *easier* than GeV LWFA

The historical context actually supports this well. The first quasi-monoenergetic LWFA results (2004, "dream beam" papers) hit ~ 70 – 200 MeV, but the experiments *before* that — and many low-power experiments since — produced broad, thermal/exponential spectra extending from near-zero up to a few MeV, often as an unwanted byproduct of "weakly relativistic" wakefields driven by modest $a_0 \sim 1$ lasers. Your 0.2 MeV target sits squarely in that regime — it's not pushing a frontier, it's deliberately operating in the *low-energy tail* that early/modest LWFA naturally produces.

Key physical point: at 200 keV, the dephasing length needed is only **microns** — comparable to a single plasma wavelength. This is actually a *feature*: a short dephasing length naturally self-limits the energy gain without requiring density tapering or staging.

Suggested parameters

Plasma:

- Density $n_e \sim 10^{19}$ – 10^{20} cm^{-3} (higher than typical multi-GeV LWFA's 10^{17} – 10^{18} cm^{-3})
- $\lambda_p \sim 3$ – $10 \text{ }\mu\text{m}$ at these densities — comparable to the laser wavelength, giving a compact "bubble" and intrinsically short acceleration length
- Wave-breaking field $E_{WB} \sim 3$ – 10 GV/m at this density range

Laser (using your stated 35 fs / 35 mJ $\rightarrow$ 1 TW peak power):

- Focus to $w_0 \sim 3$ – $5 \text{ }\mu\text{m}$ spot $\rightarrow$ intensity $\sim 10^{18} \text{ W/cm}^2 \rightarrow a_0 \sim 1$ – 1.3
- This is the **weakly relativistic regime** — exactly the a_0 used in 1990s–early 2000s self-modulated LWFA experiments that produced broad sub-MeV to few-MeV spectra
- No need for the multi-Joule, low-density, meter-scale setups used for 10 GeV records

Acceleration length: essentially the plasma source length itself — tens of μm to $\sim 100 \text{ }\mu\text{m}$, i.e. a **sub-mm gas jet**, not a cm-scale capillary or channel.

Practical implication

This injector is technologically *much* simpler than a modern high-energy LWFA stage: short gas jet (not a guided channel), modest laser intensity, no plasma waveguide needed, and the short dephasing length gives natural energy selection near 0.2 MeV without active control. The main engineering challenges shift from "how to reach high energy" to "how to control shot-to-shot reproducibility and emittance" of this low-energy, broad-spectrum output — which then needs to be captured efficiently into the twin-axis SRF cavity acceptance.

Caveat: broad/thermal spectra from this regime mean significant energy spread and divergence at extraction — likely the dominant challenge for efficient capture into the SC cavity, more so than generating the 0.2 MeV electrons themselves.

Question:

Thank you! Could you please give me one-two most relevant references from the past?

Response:

Two highly relevant references from the early LWFA era, matching the regime you'd be operating in ($a_0 \sim 1$, sub-MeV to few-MeV broad-spectrum electrons):

1. **Modena, A. et al., "Electron acceleration from the breaking of relativistic plasma waves," *Nature* 377, 606–608 (1995)** — one of the earliest demonstrations of self-modulated LWFA producing energetic electrons via wavebreaking near $a_0 \sim 1$, the regime your 35 mJ/35 fs laser naturally falls into.
2. **Malka, V. et al., "Electron acceleration by a wake field forced by an intense ultrashort laser pulse," *Science* 298, 1596–1600 (2002)** — demonstrates forced LWFA with a pulse length comparable to the plasma period (similar to your 35 fs at $n_e \sim 10^{19} \text{ cm}^{-3}$ where $\lambda_p \sim 10 \text{ } \mu\text{m}$), producing relativistic electrons from a compact, short-pulse-driven wake — closer to your target regime than the later 2004 "monoenergetic beam" results.

Both predate the 2004 breakthrough (Faure, Geddes, Mangles — all *Nature* 431, 2004) that pushed toward higher energies and narrower spectra; for your 0.2 MeV injector, the 1995–2002 broad-spectrum, modest- a_0 regime is the more directly relevant precedent.

Question:

Could you please think if 0.2 MeV energy as we assume to be coming from LWFA injector, is perhaps too low, due to dephasing of such electrons with laser bubble? What energy (the low, the better) should we consider for LWFA injector? Perhaps 0.5 MeV?

Response:

Your instinct is correct — **0.2 MeV is likely too low, and 0.5 MeV (or even ~1 MeV) is a better choice**, but for two distinct reasons, not just one.

1. Velocity mismatch with the wake (the "dephasing" issue you raised)

At plasma densities of interest ($n_e \sim 10^{19}\text{--}10^{20} \text{ cm}^{-3}$), the wakefield's phase velocity is set by the laser group velocity: $\gamma_{\text{phase}} = \sqrt{n_{\text{cr}}/n_e}$, giving $\gamma_{\text{phase}} \sim 4\text{--}13$ ($\beta_{\text{phase}} \sim 0.97\text{--}0.997$). For an electron to be genuinely "trapped" and co-moving with the bubble for extended acceleration, it needs γ approaching γ_{phase} — i.e., several MeV.

At 0.2 MeV, $\gamma \approx 1.39$, $\beta \approx 0.70$ — well below β_{phase} . Such an electron slips relative to the bucket almost immediately. At 0.5 MeV, $\gamma \approx 1.98$, $\beta \approx 0.86$ — still below β_{phase} , but meaningfully closer.

The practical resolution: electrons born near the back of the bubble get an essentially instantaneous kick from the very strong local field (the field does the work, not the electron's initial energy). So 0.2–0.5 MeV isn't really a *stable trapped state* — it's a *snapshot during transit*. To extract electrons reproducibly at this energy, you'd need a very short plasma (sub-100 μm) with a tailored exit ramp, so electrons leave before continuing up to MeV–GeV. This is an "underdeveloped acceleration" regime — workable, but sensitive to plasma length and density profile control.

2. Velocity matching into the SRF cavity (the more decisive constraint)

This is probably the bigger practical argument for raising the injection energy. Your twin-axis Nb_3Sn cavities are presumably designed assuming $\beta \approx 1$ particles (standard for relativistic SRF structures — transit-time factors, cell lengths, RF phase synchronization all assume $v \approx c$).

- At 0.2 MeV: $\beta \approx 0.70$ — a significant mismatch; transit time through each cell would differ substantially from the $\beta=1$ design assumption
- At 0.5 MeV: $\beta \approx 0.86$ — still non-relativistic but noticeably closer
- At 1 MeV: $\beta \approx 0.94$ — much closer to the design assumption

Recommendation

0.5–1 MeV is a more defensible "sweet spot" than 0.2 MeV — not because lower is bad in principle, but because there's a practical floor set by (a) self-injection dynamics requiring β reasonably close to β_{phase} , and (b) the cavity's implicit $\beta \approx 1$ design assumption. "The lower the better" doesn't hold once you account for the cavity interface — somewhere around 0.5–1 MeV likely minimizes total system complexity (LWFA extraction + capture optics + first-cell transit-time mismatch combined), rather than 0.2 MeV.

If you want, I can extend the earlier injector parameter estimate (plasma density, a_0 , acceleration length) for a 0.5 MeV or 1 MeV target instead of 0.2 MeV.

Question:

I would think the velocity matching into SRF cavity can be solved by other methods. The present assumption is three cells of dual-axis cavities. I can change them to be three single-cells dual axis cavity, and adjust the design of the first cavity for very low beta. Assuming this, please optimize the design of LWFA and its energy based on plasma acceleration primarily. Remember that it is beneficial to have as low energy as possible, for LWFA injection, to take most advantage from the energy recovery. I.e. the ratio of the top energy (around 3 MeV in our case, and it is fixed by the selected water window wavelength) to the LWFA injection energy needs to be maximized.

Response:

Once the SRF velocity-matching constraint is removed, the optimization flips around: **0.2 MeV turns out to be close to the practical sweet spot already**, and the document's original assumption was reasonably well-chosen.

Why "lower is better" hits a wall around ~0.1–0.3 MeV

The constraint isn't dephasing in the sense I framed it before (γ_{phase} doesn't set a hard injection floor — electrons born near-rest at the bubble's back asymptotically climb toward γ_{phase} , so low-energy injection is dynamically natural). The real floor is **extraction and transport**:

- **Directionality/divergence**: at 0.1 MeV, $\beta \approx 0.55$ — for a roughly thermal initial distribution, this corresponds to an enormous divergence half-angle. Capturing such a beam with any realistic optics (solenoids, etc.) becomes the dominant loss mechanism.
- **Distinguishability from background**: below ~0.1 MeV, "accelerated" electrons blend into the thermal plasma population (eV–keV temperatures) and aren't a clean, extractable bunch.
- **Density tension**: pushing the wake's "ceiling energy" itself down to ~0.3–0.5 MeV would require n_e approaching critical density (10^{20} – 10^{21} cm⁻³), where underdense LWFA physics breaks down entirely into a different (surface/sheath acceleration) regime.

Energy-recovery ratio $R = E_{\text{top}}/E_{\text{inject}}$

E_{inject}	R	β	Comment
0.1 MeV	30	0.55	Transport essentially unworkable
0.2 MeV	15	0.70	Your original choice — near-optimal
0.3 MeV	10	0.78	More comfortable margin, R drops 33%
0.5 MeV	6	0.86	R drops by half vs 0.2 MeV

Bottom line

Going from 0.2 → 0.1 MeV doubles R (15→30) but the β drop (0.70→0.55) likely costs you more in beam loss/divergence than you gain in energy-recovery efficiency — a net negative trade. Going up to 0.3 MeV gives a more comfortable extraction margin at the cost of a 33% drop in R .

0.2 MeV remains a defensible choice — possibly even slightly on the optimistic (low) side already. If anything, I'd frame the design range as **0.15–0.3 MeV**, with the precise optimum determined by PIC simulation of the actual extraction/capture efficiency vs. energy trade-off — that efficiency curve, not the plasma physics alone, is what should set the final number.

